

DataClean AI: Quick Start Guide

Welcome to DataClean AI!

DataClean AI is a state-of-the-art, automated data cleaning and imputation engine. It uses advanced statistical analysis and XGBoost-based machine learning to automatically detect anomalies and suggest the optimal cleaning strategies for your dataset.

Step 1: Mount Your Specimen (Upload Data)

Start by navigating to the **Upload Dataset** tab. You can drag and drop your data files directly into the dropzone.

- Supported formats: CSV, Excel (.xlsx, .xls), JSON.
- Maximum file size: 100 MB.

Once uploaded, the AI will parse your data and make it available in the memory workspace.

Step 2: Review the Quality Scan

Go to the **Quality Report** tab to view the automated health analysis. The AI evaluates your data across six critical dimensions:

- **Completeness:** Detection of missing or NULL values.
- **Consistency:** Formatting irregularities and casing mismatches.
- **Accuracy:** Identification of extreme outliers.
- **Uniqueness:** Detection of duplicated records.
- **Validity:** Data type adherence.
- **Integrity:** Structural cohesion.

Step 3: Apply AI Recommendations

Navigate to **AI Recommendations**. Here, the XGBoost engine will suggest specific actions to fix detected issues.

For example, it might suggest *KNN Imputation* for missing salaries if it detects non-linear relationships, or *Winsorization* for extreme outliers to preserve data mass.

Click **Apply All Fixes** to automatically execute the AI's suggestions and instantly improve your data.

Step 4: Manual Cleaning (Optional)

If you want granular control, go to **Data Cleaning**. You can build a sequential mutation pipeline step-by-step.

Available tools include Z-Score outlier removal, Mode/Median imputation, Log Transformations, One-Hot Encoding, and more.

Step 5: Exporting the Cleaned Data

Once you are satisfied with the Before/After diffs and the new Quality Score, navigate to **Export**.

- Download your fully cleaned dataset as a CSV.
- Download a final PDF Quality Report to share with stakeholders.

Need Help? Use the built-in AI Chat Assistant in the bottom right corner of the application to ask any questions about data science techniques!